

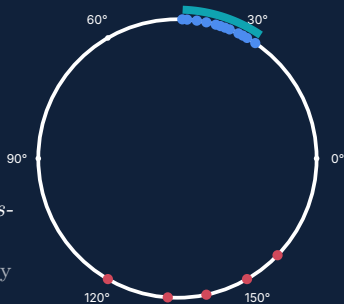
Repeated Angular Shorth Regression

Robust line fitting from the geometry of pairwise chords

Based on V. Dharavath & V. Srivastava, *Robust Linear Regression via the Repeated Angular Shorth*

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Seminar presentation · theory, geometry and simulation evidence



Roadmap

- 1 Why residual-based robust fits can fail
- 2 Chords, directions and the projective circle
- 3 The angular shorth and the *repeated* shorth
- 4 Theory: exact fit, equivariance, separation, zero influence
- 5 Why LMS is attracted by a compact cluster
- 6 Simulation evidence — and where it stops
- 7 Limitations and when to use RAShR

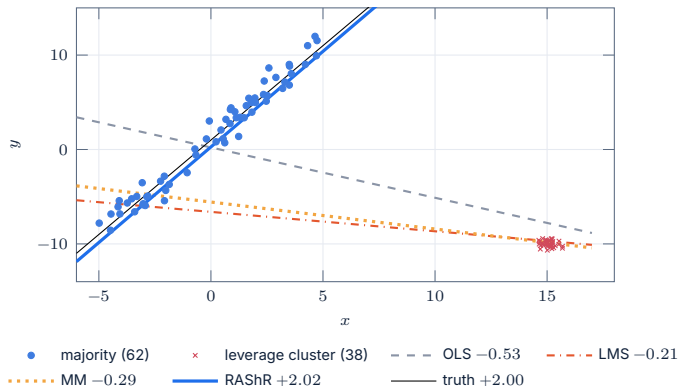
The idea in one line

Do not ask *which line has small residuals*.

Ask *which direction do pairwise chords agree on* — and measure agreement with the shortest arc that holds half of them, *twice*.

Conventions: angles of directions live in $[0, \pi)$; n observations; $\lceil \cdot \rceil$ is the ceiling. Numbers labelled *paper* are quoted from the paper; curves labelled *re-implementation* are our own simulations.

A high-breakdown fit can still follow the wrong line



$n = 100$: 62 points on $y = 1 + 2x + e$,
38 in a compact cluster at $(15, -10)$.

Slopes (paper, Fig. 1):

OLS	-0.53
LMS	-0.20
MM	-0.26
RAShR	+2.02
true slope	+2.00

Breakdown is a guarantee of boundedness

It does *not* say that the estimate follows the dominant structure.

Our data: seed 3 of the same design; slopes computed by our code (OLS -0.53, LMS -0.21, MM -0.29, RAShR +2.02).

Vocabulary, built up in order

Definition (Outlier vs. contamination)

An *outlier* is one unusual observation. *Contamination* is any set of observations not following the main pattern — possibly a large set.

Definition (Coherent contamination)

Contaminated points related to each other: a compact cluster, or a second line.

Definition (High leverage)

x_i far from the bulk of the x -values; such points pull hard on the slope.

Definition (Breakdown point)

The fraction of observations that can be replaced before the estimate can be driven arbitrarily far away.

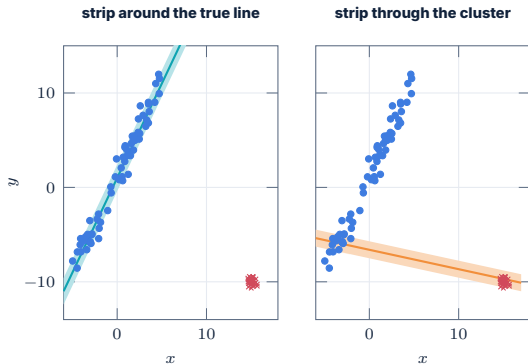
Definition (Residual-based method)

Judges candidate lines by residuals $y_i - \hat{y}_i$: LS, LMS, LTS, MM, ...

Definition (Chord)

The segment joining two observations. It has a *direction*; it is not a regression line.

Why a residual criterion can be attracted by a compact cluster



Definition (Strip width)

$W(\ell) = \inf\{w : \Pr(|Y - \ell(X)| \leq w) \geq \frac{1}{2}\}$. LMS minimises W .

- Around the truth, W must absorb the noise of the majority.
- A line through the cluster only needs a little extra majority mass.
- As contamination $\uparrow \frac{1}{2}$, that width $\rightarrow 0$.

[paper: Proposition 3 — population LMS only]

Related work as one story: pairwise slopes $\rightarrow$ pairwise directions

Method	geometric object	known behaviour	link to RAShR
Theil–Sen	median of all $\binom{n}{2}$ pairwise slopes	breakdown $\approx 29\%$	pairwise information
Siegel’s repeated median	median over i of med_j of slopes	breakdown 50%	“repeat” structure
LMS / LTS	narrowest strip / trimmed residuals	a coherent cluster can win (Prop. 3 for LMS)	the contrast case
MM	robust start + M-step (bisquare)	efficient; inherits its start	efficiency benchmark
RANSAC	2-point models, count inliers	strong on some geometries	random vs. deterministic
Hough transform	votes in parameter space	mode seeking	concentration idea
Shorth	shortest interval with half the data	robust mode / location	building block

Bridge. Keep the pairwise-geometric idea of the repeated median; replace slope-and-median aggregation by *angular concentration* through the shorth.

The central shift, and what the paper contributes

Residual-based view

Which line produces small residuals?

a line is *judged* by how well it fits a subset of points

Direction-based view

Which direction receives the strongest agreement from pairwise chords?

a line is *read off* from where the chords concentrate

Contributions

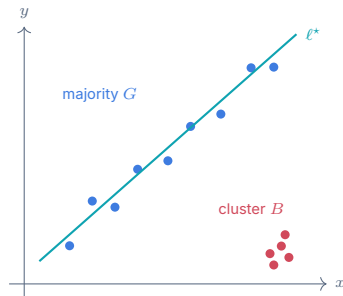
1. RAShR: a deterministic estimator of $O(n^2 \log n)$ time and $O(n)$ memory.
2. Exact fit, equivariance, and a *separation* theorem: zero influence of separated contamination (Thm 1).
3. A population result explaining why LMS is attracted by a point-mass cluster (Prop. 3).
4. A simulation study identifying a *finite* regime, $\approx 35\text{--}46\%$ coherent contamination, where RAShR keeps the majority line.
5. Honest limitations: MM is better on clean data; failure when the cluster sits along the line's extension.

Setting and notation

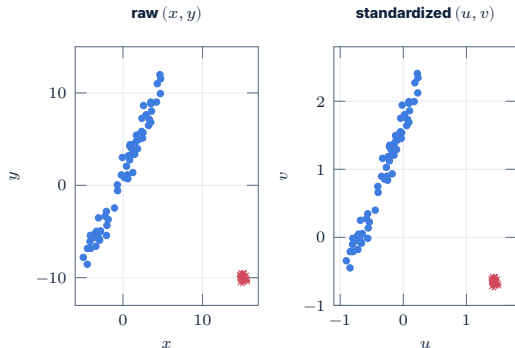
Model. Observations $(x_i, y_i)_{i=1}^n$ with majority $y_i = \alpha^* + \beta^* x_i + e_i$ on an index set G , $|G| > n/2$, and arbitrary contamination on $B = G^c$.

Goal. Estimate (α^*, β^*) although B may be large and coherent.

Symbol	Meaning
u_i, v_i	robustly standardized x_i, y_i
s_x, s_y	$1.4826 \cdot \text{MAD}$ of x , of y
φ_{ij}	direction of the chord $i \rightarrow j$, in $[0, \pi)$
d_π	distance on the projective circle
$\text{ASh}_h(A), w_h(A)$	angular shorth and its width
θ_i, w_i	local direction / width at anchor i
$\hat{\theta}$	outer (repeated) shorth



Step 1 — angles depend on units, so standardize robustly



Definition (Robust standardization) [paper Eq. (2)]

$$u_i = \frac{x_i - \text{med}(x)}{s_x}$$

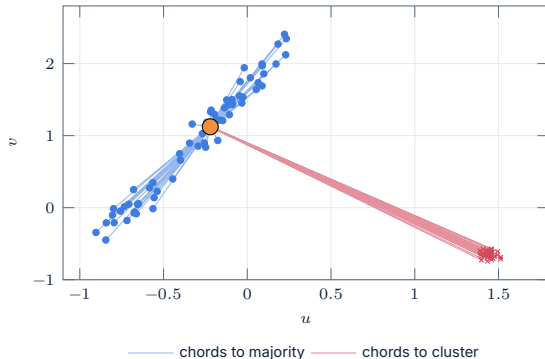
$$v_i = \frac{y_i - \text{med}(y)}{s_y}$$

with s_x, s_y equal to 1.4826 times the MAD of the coordinate. If a MAD is 0, the standard deviation of that coordinate is used.

1.4826 makes the MAD consistent for the normal standard deviation. [paper: §3.1]

On our example: $s_x = 8.553$, $s_y = 7.188$.

Step 2 — a chord is a vector; its direction is atan2

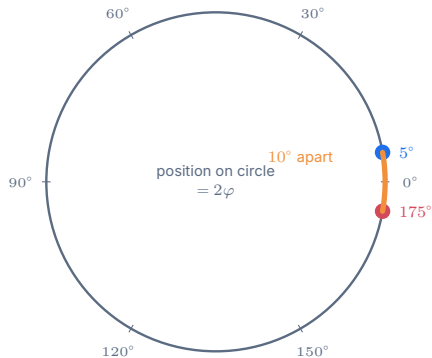


Definition (Chord direction) [Eq. (3)]

$$\varphi_{ij} = \text{atan2}(v_j - v_i, u_j - u_i) \bmod \pi$$

- Uses the vector $(\Delta u, \Delta v)$, not the ratio $\Delta v / \Delta u$: a nearly vertical chord causes no trouble.
- From an anchor i in the majority, chords to other **majority** points fan out around the true direction; chords to the **cluster** all share one very different direction.

Step 3 — directions live on the projective circle



A regression line has no orientation: φ and $\varphi + \pi$ describe the same line ($20^\circ \equiv 200^\circ$). Directions are therefore represented on the *projective circle*, identified with $[0, \pi)$ with the endpoints glued.

Definition (Projective distance)

[Eq. (4)]

$$d_\pi(a, b) = \min\{|a - b|, \pi - |a - b|\}$$

Angles near 0 and near π are *close*: 5° and 175° are only 10° apart, although $|5^\circ - 175^\circ| = 170^\circ$ on a plain number line.

Drawing convention (all deliverables): direction φ is plotted at polar angle 2φ , so the seam $0 \equiv \pi$ closes up.

The angular shorth

Definition (Angular shorth)

[§3.2]

Let $A = \{a_1, \dots, a_m\} \subset [0, \pi)$ and $1 \leq h \leq m$. $\text{ASh}_h(A)$ is the *midpoint of the shortest arc* of the projective circle that contains at least h elements of A ; its width is $w_h(A)$. Default: $h = \lceil m/2 \rceil$, written $\text{ASh}(A)$.

Computation [paper: Eq. (5)]

1. Sort $a_{(1)} \leq \dots \leq a_{(m)}$.
2. Append a copy shifted by π (arcs may cross the seam $0 \equiv \pi$).
3. Among windows of h consecutive values pick the narrowest: $r^* = \arg \min_r [a_{(r+h-1)} - a_{(r)}]$.
4. $\text{ASh}_h(A) = \frac{1}{2}(a_{(r^*)} + a_{(r^*+h-1)}) \bmod \pi$.

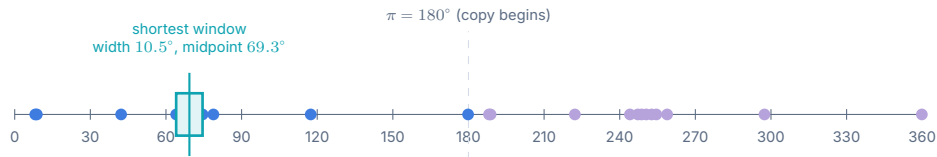
Ties: the window with the smallest left endpoint is used.

Shorth $\neq$ median angle

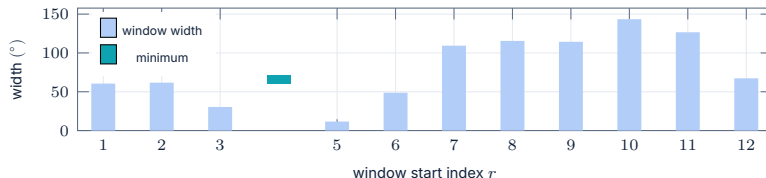
The median is the middle-ranked angle. The shorth is the centre of the *most concentrated half*. A few extreme angles — or a second, smaller, cluster of angles — cannot move it as long as the dominant concentration holds.

The angular shorth, worked on 12 angles

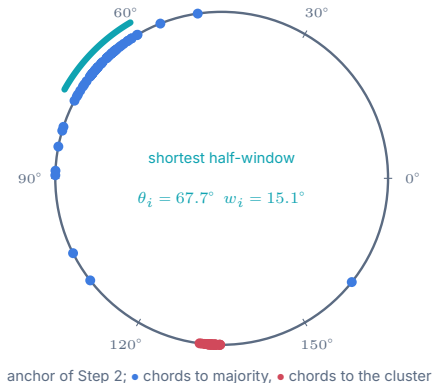
$(h = \lceil 12/2 \rceil = 6)$



• the 12 angles • copies shifted by π



Step 4 — the local direction at one anchor



Definition (Local direction and width)

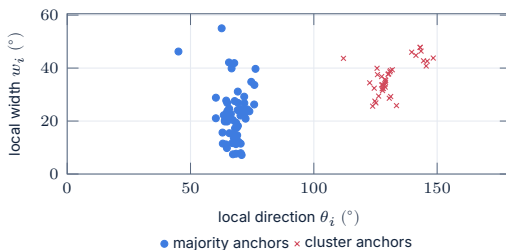
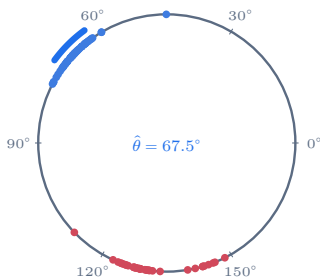
[§3.3]

For anchor i let $\Phi_i = \{\varphi_{ij} : j \neq i\}$ ($n - 1$ chord directions) and

$$\theta_i = \text{ASh}_{h_L}(\Phi_i), \quad w_i = w_{h_L}(\Phi_i), \quad h_L = \lceil \frac{n-1}{2} \rceil.$$

- w_i small: strong directional agreement seen from anchor i .
- w_i large: no clear direction from i .
- A **majority** anchor sees most majority chords near the true direction; a **cluster** anchor sees its own cluster in every direction.

Step 5 — repeat: the same shorth applied to the n local directions



Definition (Repeated angular shorth)

[Eq. (7)–(8)]

$$\hat{\theta} = \text{ASh}_{h_O} \{\theta_1, \dots, \theta_n\}, \quad h_O = \lceil n/2 \rceil.$$

The large, tight group of majority-anchor directions wins; cluster anchors form a smaller group elsewhere.

Step 6 — back to the original scale: slope and intercept

Definition (RAShR line)

[Eq. (9)–(10)]

$$\hat{\beta} = \frac{s_y}{s_x} \tan \hat{\theta}, \quad \hat{\alpha} = \text{med}_i (y_i - \hat{\beta} x_i)$$

$$\hat{y} = \hat{\alpha} + \hat{\beta} x$$

- Standardization maps slope β to $\tan \theta = \beta s_x / s_y$; undo it.
- Valid for $\hat{\theta} \in [0, \pi)$ away from $\pi/2$ (vertical).
- Intercept: every point votes $y_i - \hat{\beta} x_i$; take the median.

On the example of Fig. 1

$$\hat{\theta} = 67.5^\circ, \quad \frac{s_y}{s_x} = \frac{7.188}{8.553} \Rightarrow \hat{\beta} = 2.024$$

The intercept is less protected

$\hat{\alpha} = 0.271$ here, while $\alpha^* = 1$: the median in Eq. (10) uses *all* points, so a 38% cluster can displace the intercept even when the slope is right. The paper evaluates the *slope* error $|\hat{\beta} - \beta^*|$.

Algorithm and cost

Algorithm 1 — RAShR

[paper Alg. 1]

Require: $(x_i, y_i)_{i=1}^n$, fraction $q = \frac{1}{2}$

- 1: standardize: u_i, v_i ▷ Eq. (2)
- 2: **for** $i = 1, \dots, n$ **do**
- 3: $\varphi_{ij} \leftarrow \text{atan2}(v_j - v_i, u_j - u_i) \bmod \pi, j \neq i$
- 4: $(\theta_i, w_i) \leftarrow \text{ASh}_{h_L} \{\varphi_{ij}\}$ ▷ sort, double, slide
- 5: **end for**
- 6: $\hat{\theta} \leftarrow \text{ASh}_{h_O} \{\theta_i\}$
- 7: $\hat{\beta} \leftarrow (s_y/s_x) \tan \hat{\theta}; \hat{\alpha} \leftarrow \text{med}_i(y_i - \hat{\beta}x_i)$
- 8: **return** $(\hat{\alpha}, \hat{\beta})$

Cost

[§3.4]

n anchors $\times$ sorting $n - 1$ angles
 $\Rightarrow O(n^2 \log n)$ time; one anchor
 at a time $\Rightarrow O(n)$ extra memory.

Same order as a direct repeated median;
 cheaper than the $O(n^3)$ exhaustive
 elemental-subset search used for
 LMS/LTS in the paper's
 implementations.

Deterministic: no random sampling;
 ties go to the smallest left endpoint.

Proposition 1 and Corollary 1 — exact fit

Proposition 1 (Exact fit)

[§4.1]

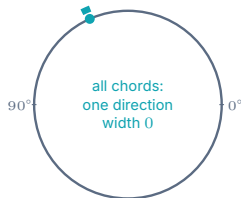
If $y_i = \alpha^* + \beta^* x_i$ for all i with distinct x_i , then $(\hat{\alpha}, \hat{\beta}) = (\alpha^*, \beta^*)$.

Proof. After standardization all points still lie on a line, so every chord has the same direction $\theta^* = \arctan(\beta^* s_x / s_y)$. For each anchor the shortest-half arc has width 0 and is centred at θ^* ; the same holds at the outer stage, so $\hat{\theta} = \theta^*$ and $\hat{\beta} = (s_y / s_x) \tan \theta^* = \beta^*$. Finally $y_i - \beta^* x_i = \alpha^*$ for every i , so $\hat{\alpha} = \alpha^*$. $\square$

Corollary 1 (Exact fit under contamination)

[§4.2]

If $y_i = \alpha^* + \beta^* x_i$ on G (distinct x_i), no point of B lies on this line, and $|B| < \lceil \frac{n-1}{2} \rceil$, then $(\hat{\alpha}, \hat{\beta}) = (\alpha^*, \beta^*)$ wherever B is placed.



geometry: a zero-width window

Proposition 2 — translation and positive-scale equivariance

Proposition 2

[§4.1]

Let $x'_i = a_x + b_x x_i$, $y'_i = a_y + b_y y_i$ with $b_x, b_y > 0$. Then

$$\hat{\beta}' = \frac{b_y}{b_x} \hat{\beta}, \quad \hat{\alpha}' = a_y + b_y \hat{\alpha} - \hat{\beta}' a_x,$$

and the estimate is invariant to permutations of the observations.

Proof. Medians are equivariant under translation and positive rescaling, MADs under positive rescaling. Hence the standardized data (u_i, v_i) are *unchanged*, all chords, shorths and $\hat{\theta}$ are unchanged, and only the back-transformation $\hat{\beta} = (s_y/s_x) \tan \hat{\theta}$ picks up the factor b_y/b_x . $\square$

What is not claimed

Not *fully* affine equivariant: $y \mapsto y + \gamma x$ changes the scale s_y and the standardized cloud is not simply rotated, so the chord directions are not shifted rigidly by a constant.

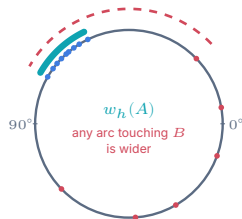
Lemma 1 — adding separated angles does not move the shorth

Lemma 1 (Separation)

[§4.2]

Let A, B be finite multisets on the projective circle, $h \leq |A|$. Suppose the shortest arc containing h elements of A is unique, with width $w_h(A)$, and that every arc containing $\geq h$ elements of $A \cup B$ and at least one element of B has width $> w_h(A)$. Then $\text{ASh}_h(A \cup B) = \text{ASh}_h(A)$.

Proof. Let I be the unique shortest h -arc of A . If I contained a point of B it would be an arc with $\geq h$ elements of $A \cup B$ meeting B and of width $w_h(A)$, contradicting the hypothesis; so I is still feasible for $A \cup B$. Any feasible arc meeting B is strictly wider; one using only A is not narrower than I , and equally wide only if it equals I . So I is the unique shortest feasible arc. $\square$



A : dominant group; B : separated angles

Separation conditions at the two stages

Let G be the majority, $|G| > n/2$, $B = G^c$. For $i \in G$ let $\Phi_i^G = \{\varphi_{ij} : j \in G \setminus \{i\}\}$. The two shortest-half sizes are

$$h_L = \left\lceil \frac{n-1}{2} \right\rceil, \quad h_O = \left\lceil \frac{n}{2} \right\rceil.$$

Since $|G| > n/2$: $|G| - 1 \geq \lfloor n/2 \rfloor \geq h_L$ and $|G| \geq h_O$ — both shorths *can* be built from majority observations alone.

(L) (Local separation)

For every $i \in G$, Lemma 1 applies with $A = \Phi_i^G$, $B = \{\varphi_{ij} : j \in B\}$, $h = h_L$.

(O) (Outer separation)

Lemma 1 applies to the local directions with $A = \{\theta_i : i \in G\}$, $B = \{\theta_i : i \in B\}$, $h = h_O$.

In words

(L): at every majority anchor, the shortest-half interval of majority–majority chords is narrower than any equally large interval involving a contaminated chord. (O): at the second stage, the concentrated majority-anchor directions form a narrower half-sample interval than any interval involving a contaminated-anchor direction.

Theorem 1 — zero influence of separated contamination

Theorem 1

[§4.2, Eq. (11)]

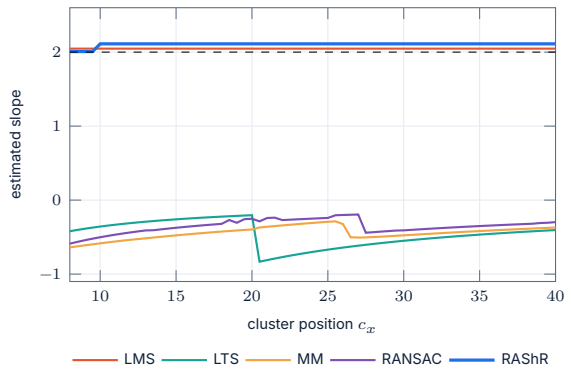
Suppose (L) and (O) hold. Then

$$\hat{\theta} = \text{ASh}_{h_O} \left\{ \text{ASh}_{h_L}(\Phi_i^G) : i \in G \right\}.$$

So, once the standardization scales are fixed, $\hat{\theta}$ and hence $\hat{\beta}$ are determined by the majority observations only: moving, adding or removing contaminated observations does not change $\hat{\beta}$, provided (L), (O) and $|G| > n/2$ continue to hold and the marginal medians and MADs used in Eq. (2) remain unchanged.

Proof. For $i \in G$, (L) and Lemma 1 give $\theta_i = \text{ASh}_{h_L}(\Phi_i) = \text{ASh}_{h_L}(\Phi_i^G)$. Hence the local direction at every majority anchor depends on majority-majority chords only. Applying (O) and Lemma 1 once more gives $\hat{\theta} = \text{ASh}_{h_O}\{\theta_i : i \in G\}$, which is the claim. $\square$

Reading Theorem 1 — the horizontal cluster-movement experiment



A fixed 36-point cluster is moved horizontally; the majority sample is held fixed.

Paper: the RAShR slope becomes *exactly constant at 2.058* once the cluster is sufficiently separated — the mechanism of Theorem 1.

Here (our sample): the same plateau, at a different value because the majority sample differs; residual-based fits jump toward the cluster at different positions.

Caveat

Theorem 1 needs s_x, s_y and the medians in Eq. (2) to stay fixed; contamination still enters them.

Proposition 3 — LMS is attracted by a point-mass cluster

Model. $(X, Y) \sim (1 - \varepsilon)P_G + \varepsilon\delta_c$, where under P_G : $Y = \alpha^* + \beta^*X + e$, $e \sim N(0, \sigma^2) \perp X$, and $c = (c_x, c_y)$ lies at vertical distance $\Delta = |c_y - \alpha^* - \beta^*c_x| > 0$ from the majority line.

$$W(\ell) = \inf\{w : \Pr(|Y - \ell(X)| \leq w) \geq \tfrac{1}{2}\}.$$

Proposition 3

[§4.3]

There is $\varepsilon_0 < \frac{1}{2}$ such that for every $\varepsilon \in (\varepsilon_0, \frac{1}{2})$ every population LMS solution ℓ satisfies $|c_y - \ell(c_x)| \leq W(\ell)$, with $W(\ell) \rightarrow 0$ as $\varepsilon \uparrow \frac{1}{2}$.

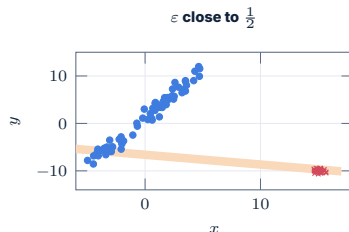
Consequently ℓ^* is not an LMS solution, and every LMS solution passes arbitrarily close to c .

Scope

Proved for *population LMS*, asymptotically in $\varepsilon \uparrow \frac{1}{2}$. For LTS and MM, whose scale or subset criteria behave similarly, no theorem is claimed: their behaviour is studied empirically (Section 5). Finite-sample experiments show *how early* the attraction becomes visible.

Proposition 3 — why it holds

1. Let w_ε solve $(1 - \varepsilon) \Pr(|e| \leq w_\varepsilon) = \frac{1}{2}$. For $\varepsilon \geq \frac{1}{4}$:
 $w_\varepsilon \geq w_{1/4} > 0$.
2. Any strip that *excludes* c and holds half of the mass contains only majority mass; by Anderson's inequality $\Pr(|e - t| \leq w) \leq \Pr(|e| \leq w)$, so its half-width is at least $w_\varepsilon \geq w_{1/4}$.
3. Take any non-vertical line ℓ_c through c . A strip of half-width w around it already holds the whole mass ε . It holds half of the total mass once $(1 - \varepsilon)p(w) \geq \frac{1}{2} - \varepsilon$,
 $p(w) = P_G(|Y - \ell_c(X)| \leq w) > 0$. As $\varepsilon \uparrow \frac{1}{2}$ the right-hand side $\rightarrow 0$, so $W(\ell_c) \rightarrow 0$.
4. Choose $\varepsilon_0 \geq \frac{1}{4}$ with $W(\ell_c) < \min\{w_{1/4}, \Delta\}$ on $(\varepsilon_0, \frac{1}{2})$. An LMS solution has $W(\ell) \leq W(\ell_c) < w_{1/4}$, so its optimal strip must contain c , i.e. $|c_y - \ell(c_x)| \leq W(\ell) \leq W(\ell_c) \rightarrow 0$. Because $W(\ell_c) < \Delta$, the majority line ℓ^* (at vertical distance Δ from c) cannot be a solution.



a thin strip through the cluster only has to collect a little extra majority mass

Remark 1 — why an “all chords” concentration condition is too strong

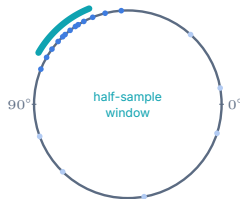
The tempting assumption

Assume every majority–majority chord lies within d_π -distance $\delta < \pi/8$ of the true direction θ^* . One can then derive $d_\pi(\hat{\theta}, \theta^*) \leq 4\delta$.

Valid mathematically, but **useless for noisy regression**: two observations with nearly equal x and a little vertical noise produce a chord with an almost arbitrary direction. Requiring *all* majority chords near θ^* is far too restrictive.

What RAShR needs instead

Concentration of a *half-sample* of chords — exactly what the shorth detects — not concentration of all of them.



Simulation design (paper, Section 5)

Data

$n = 100$, $x \sim U[-5, 5]$, $y = 1 + 2x + e$, $e \sim N(0, 1)$. A fraction f of the observations is replaced by one of three *pre-declared* patterns:

1. compact high-leverage cluster at $(15, -10)$, Gaussian spread 0.3;
2. competing line $y = 5 - 1.25x + e_c$, $e_c \sim N(0, 0.5^2)$ over the same x -range;
3. heteroscedastic majority (error sd $0.2 + 0.6|x|$) plus a competing line with noise sd 0.3.

Measures

Absolute slope error $|\hat{\beta} - 2|$; failure = $\Pr(|\hat{\beta} - 2| > 0.5)$; 60 replications.

Competitors

LMS; **LTS**: ten best elemental fits + 20 concentration steps; **MM**: LTS start, MAD scale, Tukey bisquare; **RANSAC**: two-point samples, MAD-based residual threshold, 1000 trials.

Reading our curves

The paper does not fully specify some constants (notably RANSAC's threshold). Our curves come from *our* implementation with fixed seeds; they reproduce the *pattern*, not each number. Paper values are always shown separately; see the appendix for a threshold-sensitivity study.

Standard settings: RAShR is not uniformly preferable

Paper — median $|\hat{\beta} - 2|$

setting	RAShR	MM	RANSAC
clean data	0.033	0.024	0.022
20% vertical	0.040	0.027	0.037
10% bad leverage	0.041	0.020	0.021

On clean data and in standard robust-regression settings, **MM-regression is generally the more efficient choice**. RAShR's advantage appears only when the contaminating group is both *large* and *geometrically coherent*.

Our re-implementation (100 replications each)

setting	RAShR	LMS	LTS	MM	RANSAC
clean	0.043	0.054	0.051	0.022	0.022
20% vertical	0.034	0.052	0.056	0.023	0.056
10% bad leverage	0.042	0.054	0.067	0.022	0.022
20% competing line	0.045	0.070	0.060	0.032	0.061

Same ordering as the paper: MM (and a tuned RANSAC) win on efficiency in standard settings.

Near-half coherent contamination — what the paper reports

(a) compact leverage cluster — failure %

f	RAShR	LMS	LTS	MM	RANSAC
36%	0	2	23	23	50
40%	0	58	100	100	77
44%	60	≥ 97	≥ 97	≥ 97	≥ 97
46%	all methods fail				

44%: RAShR succeeds in 40% of samples (fails 60%);
every competitor fails in at least 97%.

(b) competing line — RAShR: no failures through 46%

f	RAShR	LMS	LTS	MM	RANSAC
46%	0	60	93	93	12

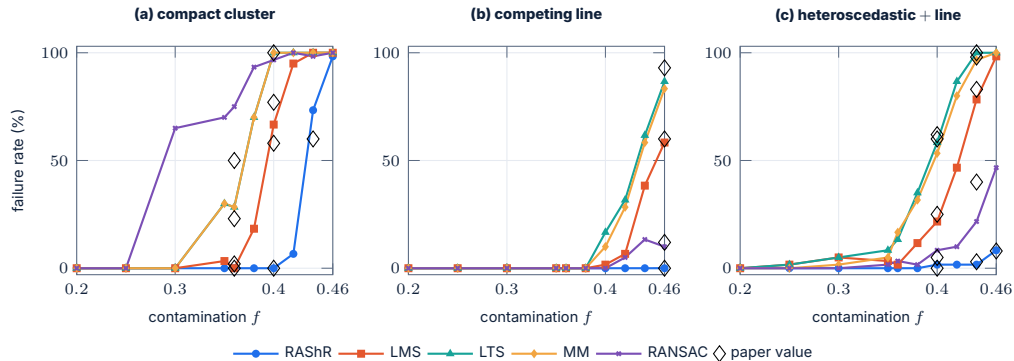
(c) heteroscedastic majority + line

f	RAShR	LMS	LTS	MM	RANSAC
40%	0	25	62	60	5
44%	3	83	100	98	40
46%	8	—	—	—	—

Not a general “near-50%” guarantee

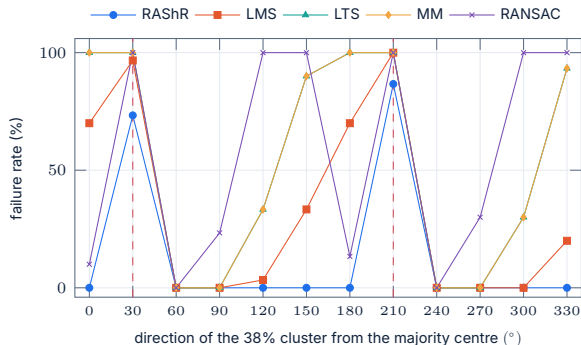
Compact cluster: all methods fail at 46%. The experiments identify a *finite regime*, ≈ 35 –46%, where RAShR can stay near the majority line after established methods have started to follow the competing structure; below $\approx 30\%$ the high-breakdown methods behave alike.

Failure rate versus contamination — our re-implementation, with the paper's points



Curves: our code, 60 replications per point, same designs as the paper. Diamonds: values reported in the paper's text (for (a) at 44%: RAShR; other competitors $\geq 97\%$ not drawn).

When the advantage disappears: cluster direction and spread



our code; distance 30 from the majority centre. Dashed red: 30° and $210^\circ \equiv 30^\circ$ (the line's extension) — here $-150^\circ \equiv 210^\circ$.

Paper. A 38% cluster in twelve directions, two distances: RAShR has no failures in ten of twelve directions. The difficult ones are $\approx 30^\circ$ and -150° , where the cluster lies near the *extension* of the majority line: chord directions overlap and separation is lost. At distance 30: RAShR fails 78% (30°) and 75% (-150°); competitors also perform poorly.

Compactness. Spread 0.05: RAShR 0 failures, LMS 25%, LTS/MM $\approx 98\%$. Spread 1 or 3: all four of LMS, LTS, MM, RAShR fail $\leq 2\%$ — not a compact coherent group any more.

Spread of the cluster — our re-implementation (failure %, compact cluster, 38%)

spread	RAShR	LMS	LTS	MM	RANSAC
0.05	0	20	92	92	92
0.3	0	18	70	70	93
1.0	0	0	0	0	97
3.0	0	0	0	0	88

Reproduces the qualitative statement of the paper: a very tight cluster fools LMS/LTS/MM, a diffuse one does not. (Paper: spread 0.05 gives LMS 25%, LTS/MM $\approx$ 98%, RAShR 0.)

Geometry of the effect

A very tight cluster is almost a point mass: a thin strip through it (LMS) or a high-breakdown scale (MM) latches onto it, while *its chords point everywhere* and so never form a competing half-window of directions.

RANSAC at spread ≥ 1

Our RANSAC with the sklearn-style threshold $\text{MAD}(y)$ is poor here; the paper reports no RANSAC numbers for this experiment. See the appendix on threshold sensitivity.

Strengths and limitations — side by side

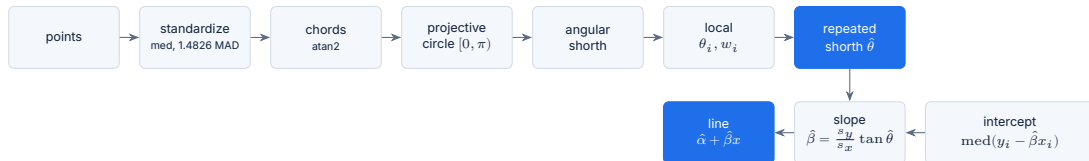
Where RAShR helps

- A *large, compact, coherent* contaminating group of about 35–46%.
- Competing structure that pulls residual criteria (LMS, LTS, MM; RANSAC for some designs).
- Deterministic output, $O(n^2 \log n)$ time, $O(n)$ memory.
- Exact fit and zero influence of separated contamination (Prop. 1, Cor. 1, Thm 1).

Where it does not

- Clean data, low contamination, vertical outliers, bad leverage: **MM** / **RANSAC** have smaller median errors.
- Cluster near the *extension* of the majority line ($\approx 30^\circ, -150^\circ$): separation is lost.
- Diffuse (non-compact) clusters: no advantage.
- Compact cluster at 46%: every method fails.
- Not fully affine equivariant; s_x, s_y and the intercept use all data.
- Prop. 3 is a population-LMS statement only.

Summary: the geometry in one chain



Theory

exact fit, equivariance, zero influence under (L),(O); Prop. 3 for LMS.

Evidence

finite regime 35–46%; compact, separated clusters; MM wins when data are clean.

Use it when

a large coherent group competes with the majority and a deterministic fit is wanted.

References (as cited in the paper) and package contents

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This package

Manim video (60 fps, narrated)
 · PowerPoint (geometry-first) ·
 interactive HTML slides · this
 Beamer deck · Streamlit lab with
 a simulation sandbox · note-
 books. All use one shared es-
 timator, one notation and the
 same paper numbers.

Appendix A — Corollary 1: proof

Corollary 1

[§4.2]

If $y_i = \alpha^* + \beta^* x_i$ on G (distinct x_i), no point of B lies on the line, and $|B| < \lceil \frac{n-1}{2} \rceil$, then $(\hat{\alpha}, \hat{\beta}) = (\alpha^*, \beta^*)$ regardless of the positions of B .

Proof. At a majority anchor every chord to another majority point has the same direction θ^* : a zero-width interval with $|G| - 1 \geq h_L$ elements, hence feasible. A chord from a majority anchor to a contaminated point has direction θ^* only if that point lies on the majority line, which is excluded. Any zero-width interval at another direction contains at most $|B| < h_L$ contaminated chords, so it cannot meet the half-sample requirement. Hence $\theta_i = \theta^*$ for all $i \in G$. At the outer stage at least $|G| \geq h_O$ local directions equal θ^* while fewer than h_O arise from contaminated anchors, so these cannot form a competing zero-width interval: $\hat{\theta} = \theta^*$ and $\hat{\beta} = \beta^*$. Finally more than half of the residuals $y_i - \hat{\beta}x_i$ equal α^* ($i \in G$), so their median is α^* . $\square$

Appendix B — how the numbers in this package relate to the paper

Quantity	Paper	Our implementation
Fig. 1 slopes (RAShR / OLS / LMS / MM)	+2.02 / -0.53 / -0.20 / -0.26	+2.02 / -0.53 / -0.21 / -0.29 (seed 3)
Compact cluster, 44%: RAShR fails	60% (succeeds 40%)	55% (60 reps, appendix C samples)
Directions at distance 30: 30° / -150°	78% / 75%	see previous slide (RAShR 73% at 30°)
Cluster-move plateau	2.058	different value (different majority sample); constant plateau reproduced
RANSAC (compact, 36%)	50% fails	depends strongly on the threshold (appendix C)

Everything is deterministic given the seeds in the code. Differences in competitor numbers stem from unspecified tuning constants, not from tuning toward the paper.

Appendix C — RANSAC threshold sensitivity (failure %, 60 replications)

residual threshold	compact cluster, $f =$					line	funnel	
	30%	36%	40%	44%	46%	46%	40%	44%
MAD(y) [default]	53	93	97	100	100	13	8	20
0.5 MAD(y)	0	0	22	100	100	22	30	67
2.0 (oracle)	0	0	13	98	100	13	27	65
2.5 (oracle)	0	0	17	100	100	3	7	38
4.0 (oracle)	0	8	93	100	100	12	8	23
RAShR (same samples)	0	0	0	55	98	2	3	0

RANSAC's outcome depends strongly on a constant the paper only calls "MAD-based". With an *oracle* threshold ($2\text{--}2.5\times$ the true noise sd) RANSAC is far stronger than with $\text{MAD}(y)$, but still fails at 44–46% compact contamination, and RAShR (same samples) keeps the majority line up to 40%. Honest conclusion: RAShR's advantage is real but its size depends on how well RANSAC is tuned.